

1. What are tokens in C?

Smallest individual units of a C program such as keywords, identifiers, constants, operators, and special symbols.

2. What are identifiers?

Names given to variables, functions, arrays, structures, etc.

3. What are escape sequences?

Special characters preceded by \ used to represent non-printable characters.

4. What is the use of \n and \t?

- \n → New line
 - \t → Horizontal tab
-

5. What are comments in C?

Statements ignored by the compiler, used for documentation.

6. Difference between single-line and multi-line comments?

- Single-line: // Comment
 - Multi-line: /* Comment */
-

7. What is an expression?

A combination of variables, constants, and operators that produces a value.

8. What is operator precedence?

The priority of operators during expression evaluation.

9. What is associativity?

The order in which operators of the same precedence are evaluated.

10. What is the ternary operator?

A shorthand for if-else.

condition ? value1 : value2;

11. Difference between pre-increment and post-increment?

- ++a → Increment first, then use value.
 - a++ → Use value first, then increment.
-

12. What is implicit type conversion?

Automatic conversion of one data type to another by the compiler.

13. What is explicit type conversion?

Manual conversion using type casting.

float x = (float)5/2;

14. What are bitwise operators?

Operators that work on individual bits:

&, |, ^, ~, <<, >>

15. What is modulus (%) operator?

Returns the remainder after division.

10 % 3 = 1

16. What is the purpose of return statement?

Returns control and optionally a value from a function.

17. What is the use of const keyword?

Makes a variable read-only after initialization.

18. Difference between character and string?

- Character: 'A'
 - String: "ABC"
-

19. What is ASCII?

American Standard Code for Information Interchange; represents characters using numeric codes.

20. What is the range of char, int, and float?

- char : -128 to 127
 - int : -2,147,483,648 to 2,147,483,647
 - float : approximately $\pm 3.4 \times 10^{38}$
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MEDIUM (21–40)

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21. What is a multidimensional array?

An array containing other arrays.

```
int arr[3][4];
```

22. How are strings stored in memory?

As a sequence of characters ending with '\0'.

23. What is string termination character?

The null character '\0'.

24. Difference between strcpy() and strncpy()?

- strcpy() copies the entire string.
 - strncpy() copies only a specified number of characters.
-

25. Difference between strcmp() and strncmp()?

- strcmp() compares complete strings.
 - strncmp() compares first n characters.
-

26. What is an array of pointers?

An array whose elements are pointers.

```
char *names[5];
```

27. What is a pointer to pointer?

A pointer that stores the address of another pointer.

```
int **ptr;
```

28. Explain pointer to array.

A pointer that points to an entire array.

```
int (*ptr)[5];
```

29. Difference between array and pointer?

Array	Pointer
Fixed memory block	Stores address
Cannot be reassigned	Can be reassigned

30. What are command-line arguments?

Values passed to a program during execution.

31. Explain argc and argv.

- argc → Number of arguments.
 - argv → Array of argument strings.
-

32. What is static variable?

Retains its value between function calls.

33. What is extern variable?

A variable declared in another file or scope.

34. What is register storage class?

Requests storage in CPU registers for faster access.

35. What is auto storage class?

Default storage class for local variables.

36. What are macros?

Preprocessor definitions replaced before compilation.

#define PI 3.14

37. Difference between macro and function?**Macro****Function**

Preprocessor replacement Executed at runtime

No type checking

Type checking available

Faster

Slight overhead

38. What is file handling in C?

Performing operations like create, read, write, and append on files.

39. What are file modes in C?

- "r" → Read
- "w" → Write
- "a" → Append
- "r+" → Read/Write
- "w+" → Read/Write (overwrite)

- "a+" → Read/Append

40. Difference between text file and binary file?

Text File	Binary File
Human readable	Machine readable
Slower	Faster
Uses character representation	Stores raw bytes

HARD (41–50)

41. What is a self-referential structure?

A structure containing a pointer to itself.

```
struct Node {  
    int data;  
    struct Node *next;  
};
```

42. What is typedef in C?

Creates an alias for an existing data type.

```
typedef unsigned int UINT;
```

43. What are enumerations (enum)?

User-defined data types consisting of named integer constants.

```
enum Day {MON, TUE, WED};
```

44. Difference between #include <file.h> and #include "file.h"?

- <file.h> → Searches system directories.
 - "file.h" → Searches current directory first.
-

45. What is conditional compilation?

Compiling selected code based on conditions.

```
#ifdef DEBUG  
printf("Debug");  
#endif
```

46. Explain preprocessing directives.

Instructions executed before compilation.

Examples:

```
#include  
#define  
#ifdef  
#ifndef  
#undef
```

47. What is segmentation fault?

A runtime error caused by accessing invalid memory locations.

48. What are command-line macros?

Macros defined during compilation.

```
gcc prog.c -DVALUE=10
```

49. What is the difference between stack memory and heap memory?

Stack	Heap
Automatic allocation	Dynamic allocation
Faster	Slower
Managed by compiler	Managed by programmer
Limited size	Larger size

50. Explain the memory layout of a C program.

1. **Code Segment** → Stores executable instructions.
2. **Data Segment** → Stores initialized global/static variables.
3. **BSS Segment** → Stores uninitialized global/static variables.
4. **Heap** → Dynamic memory (malloc, calloc).
5. **Stack** → Function calls, local variables, parameters.

High Address

Stack

Heap

BSS

Data

Code

Low Address

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